

PostureGuard

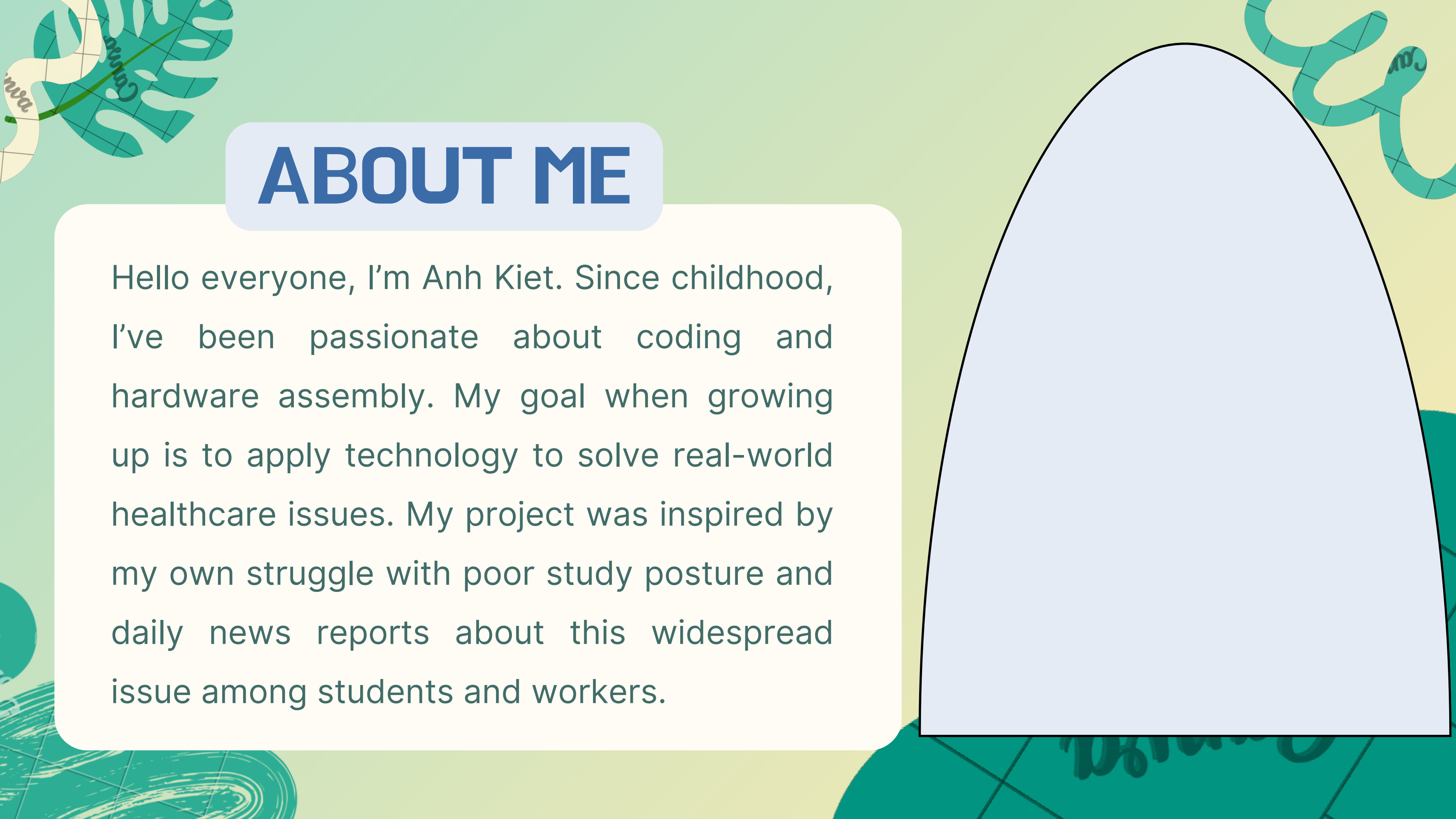


PostureGuard

How to Take Care of Your Physical Health

Presentation by Nguyen Anh Kiet



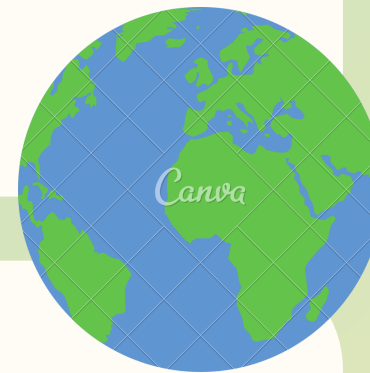


ABOUT ME

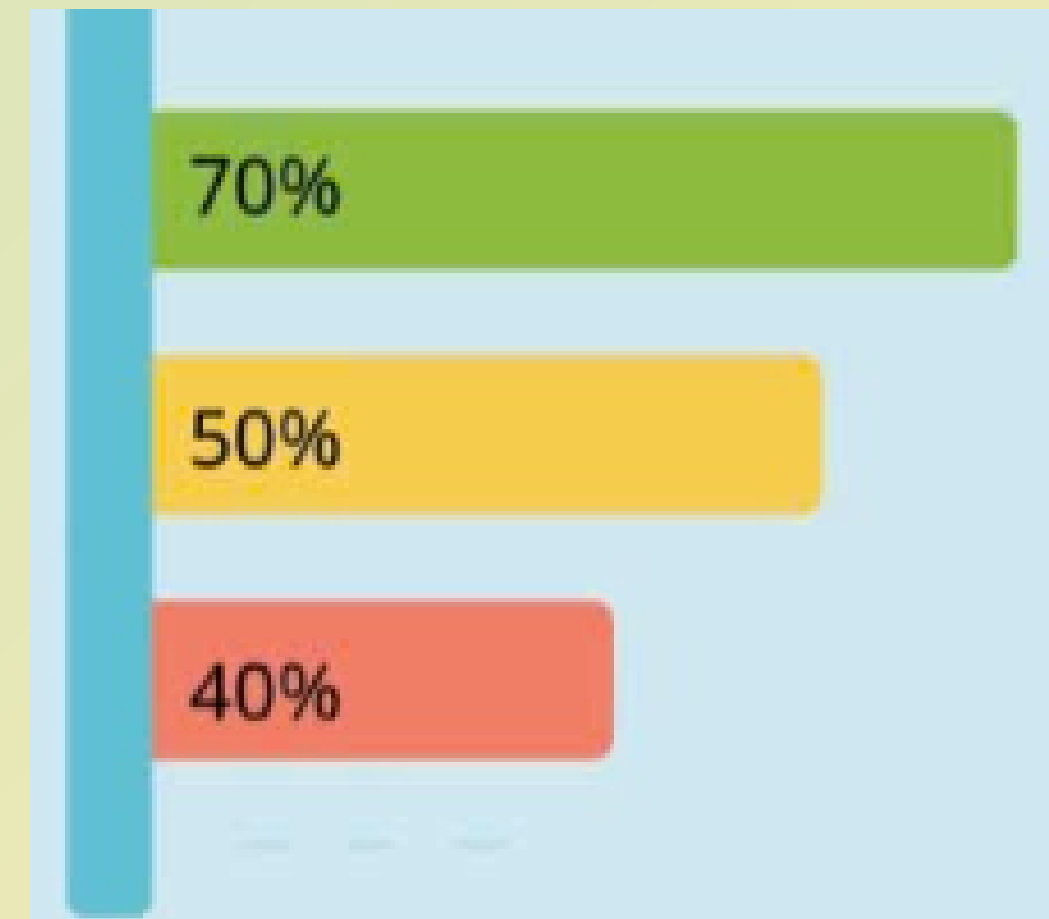
Hello everyone, I'm Anh Kiet. Since childhood, I've been passionate about coding and hardware assembly. My goal when growing up is to apply technology to solve real-world healthcare issues. My project was inspired by my own struggle with poor study posture and daily news reports about this widespread issue among students and workers.

CURRENT SITUATION

- 27 KG of pressure on the cervical spine when leaning forward at a 60 degree angle.
- Over 8 hours of daily sitting increases health risks



- 70% of office workers & 50% of students constantly sit in incorrect postures, leading to chronic back and neck pain.



ROOT CAUSE

1. Lack of Awareness

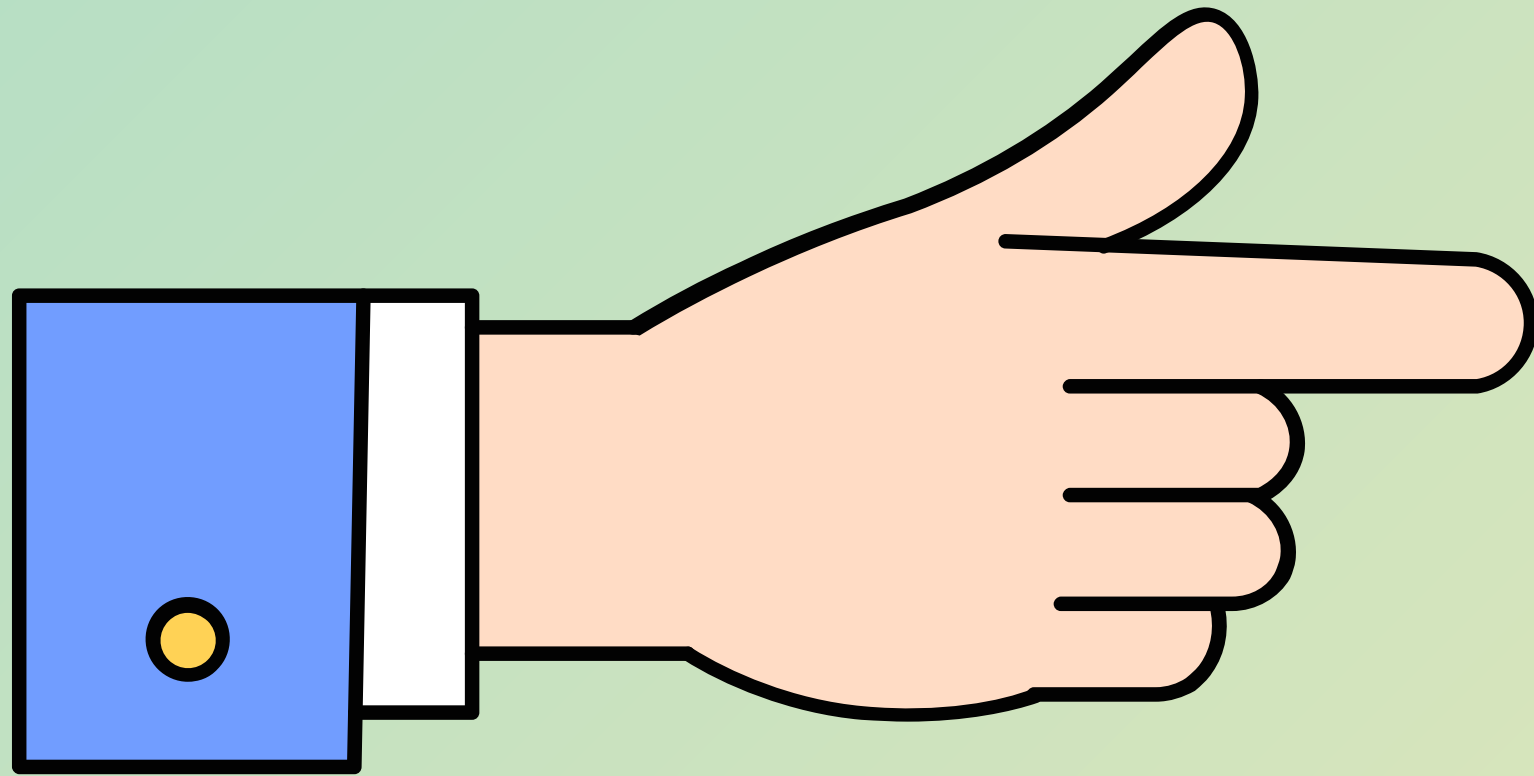
- Users are completely unconscious of their poor posture until pain occurs.

2. Prolonged Screen Time

- Heavy workloads force the body into a static position for hours without breaks.

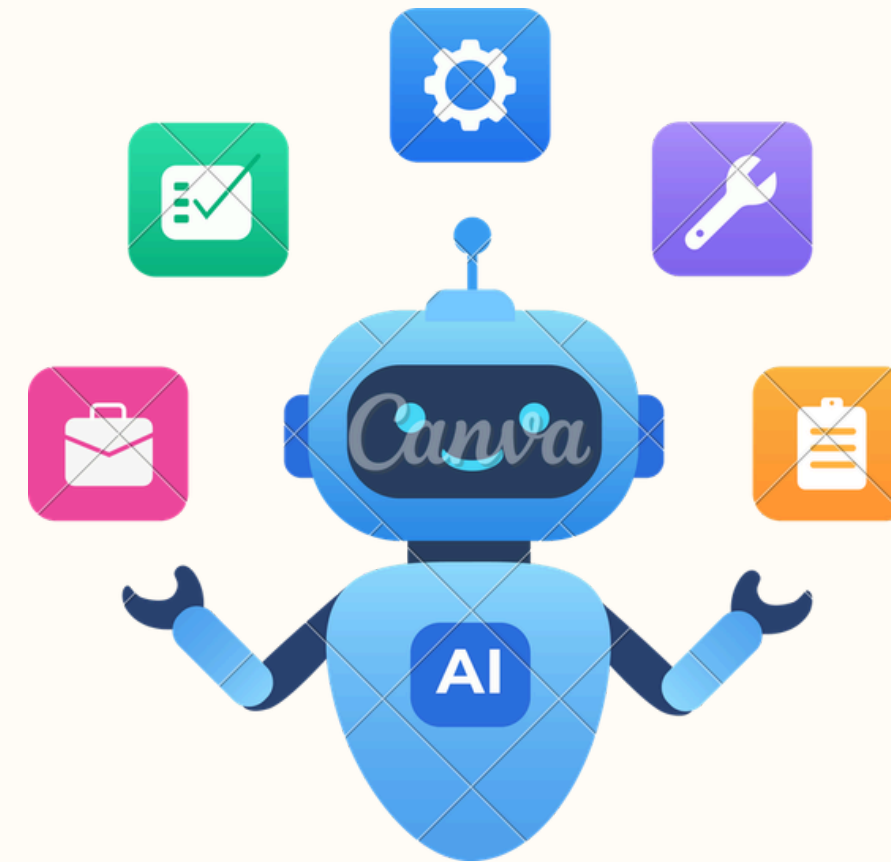
3. Ergonomic Mismatch

- Standard desks and chairs do not fit individual heights, forcing the spine to bend to adapt.



MY GOAL

Developing an intelligent, non-invasive system to protect the physical health of students and office workers.



Real-time Alert

Detect bad posture instantly using camera vision



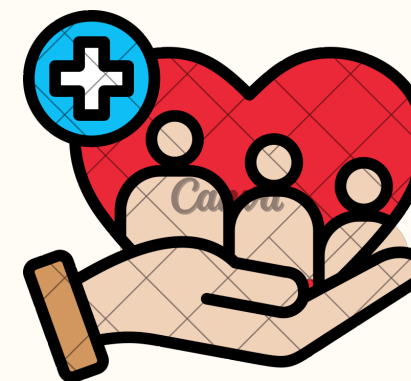
AI Data Analysis

Process posture habits through computer vision



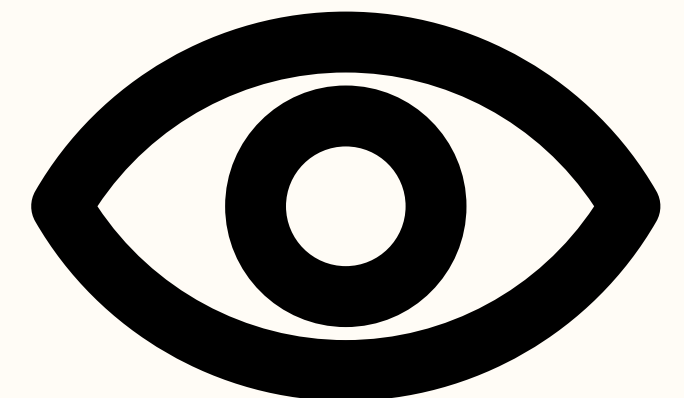
Habit Reformation

Build a long-term, self-aware routine of sitting straight



Vision Protection

Smart screen-distance monitoring



Theoretical research

1. Research bad posture risks among students/workers.
2. Hardware: Real-time distance sensor
3. Software: Camera-based computer vision models for body alignment.
4. Analyze existing products to define ErgoGuard's unique features.



Data Collection & Analysis

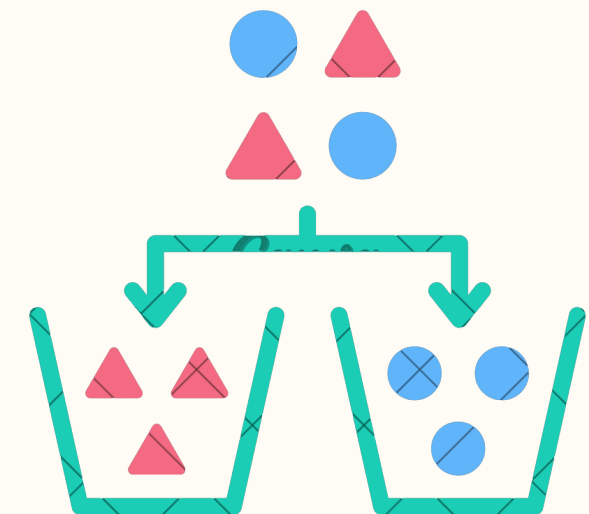
Map out circuit wiring
connecting Arduino Uno, LCD
1602, sensor, and buttons.



Capture
video/photo of real-
world sitting habits



Sort dataset into
standard categories:
Correct and Incorrect

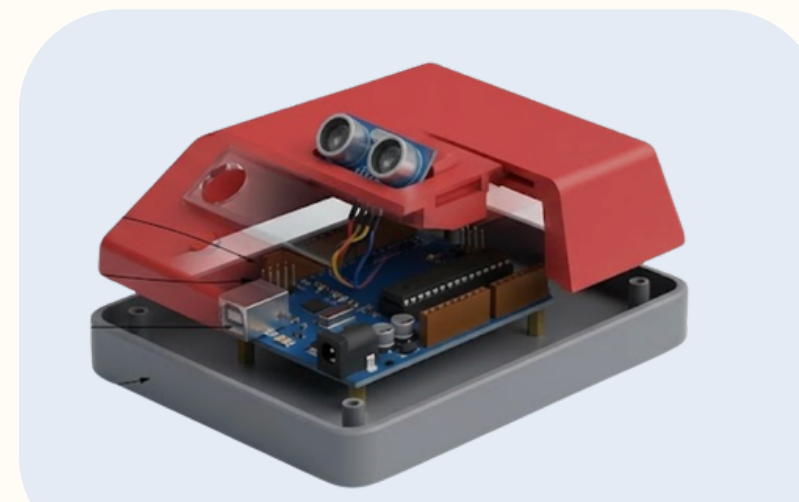


Hardware & Software Development

- Model a custom ergonomic case in Tinkercad and 3D print it.
- Code distance logic, mode-switching buttons, and LCD display outputs in C++
- Build a desktop/web app that triggers the web camera and runs the core detection model.



Software



Hardware

Conclusion

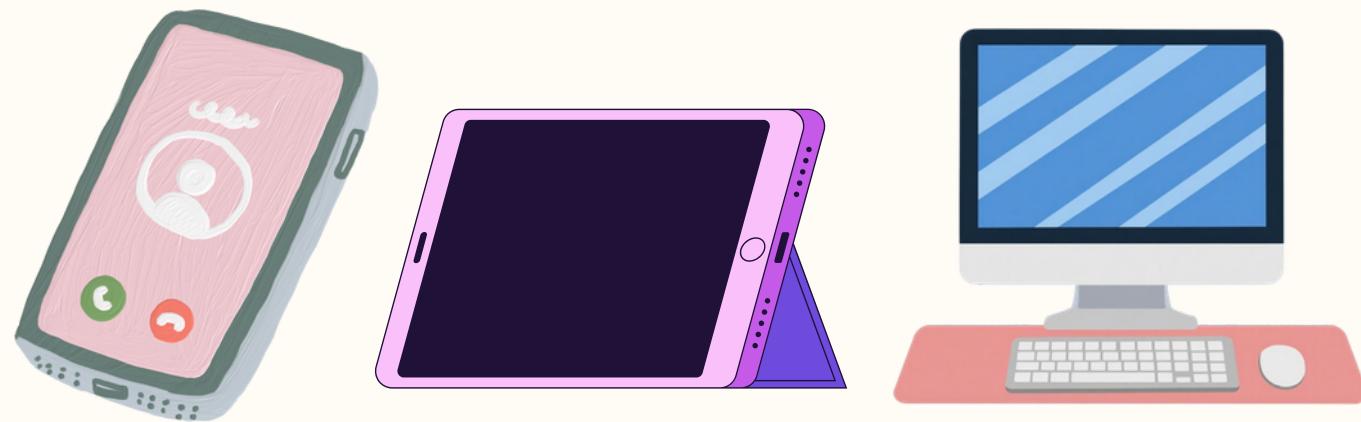
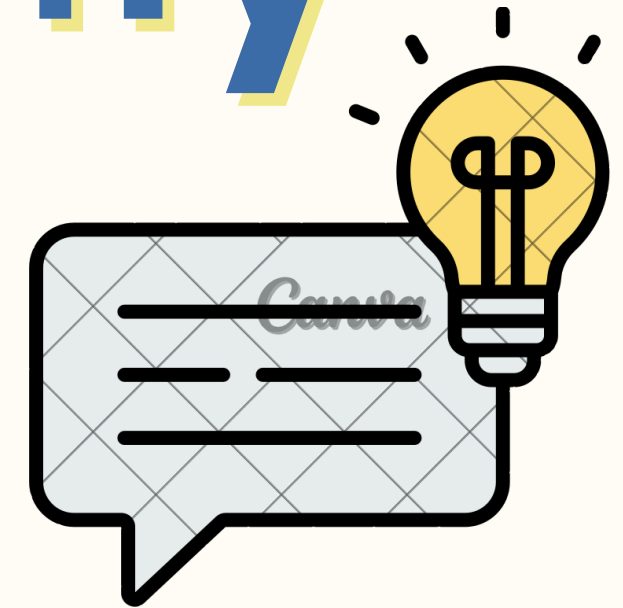
An app:

- Reduces negative impacts on eye health and the spine.
- Tracks posture and screen time to give automatic, timely reminders.
- Tracks device usage habits and suggests custom eye and spine exercises.
- Easy to use with visual health reports for tracking and adjusting habits.



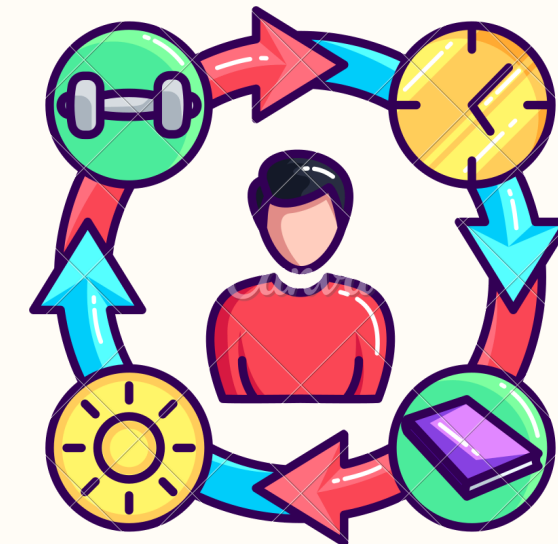
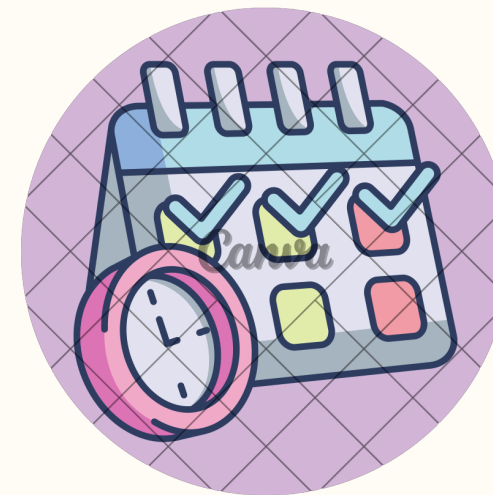
Novelty & Creativity

- AI application in vision and spine care: automatically detect bad sitting posture or excessive screen time and provide instant alerts to fix a user's habits



- Convenience and accessibility: works on common devices and requires no special equipments

- Comprehensive approach: offers personalized solutions to improve daily habits



Future development

01

**Improve
data
analysis
capabilities**



02

**Expand
health
protection
features**



03

**Cross-
platform
application**



04

**Tailor the
app for
different
ages**



05

**Comprehensive
evaluation
feature**





Thank You

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